

InnoDevs SRS DOCUMENT

SOFTWARE REQUIREMENTS SPECIFICATION(SRS)

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1.1 Introduction

The Purpose of the Multi-Language Phrasebook is to provide DUT students from different Linguistic-Backgrounds who are left at a disadvantage when it comes to Campus important resource information such as Registration, Accommodation, and Health Services because this information is mostly in English. The system aims to break down language barriers by allowing students to find translations for important campus-related information and contribute their own translations for review.

1.2 Problem Description

DUT is home to students from different linguistic backgrounds across KZN and beyond yet essential campus information covering registration, accommodation, health services and academic support is mainly conveyed in English, creating a significant language barrier for students who are more comfortable communicating in isiZulu, Sesotho, isiXhosa, or any of the other official South African languages. As a result, these students are placed at a disadvantage when trying to access and understand information that is critical to their day-to-day experience at DUT. This communication gap can lead to confusion, delays, missed deadlines, and a general sense of exclusion for students who don't have strong English proficiency.

1.3 Proposed Solution

The InnoDevs have come together and decided to develop a Multi-Language Campus Phrasebook which will bridge a gap to learners who are most comfortable in their mother tongue language from being disadvantaged when it comes to accessing different services around campus.

The Multi-Language Campus Phrasebook directly addresses this gap by providing a searchable, community-driven platform where:

- Students can quickly find translations of important campus phrases in their preferred language
- Students can contribute their own translation suggestions, helping their peers
- Administrators can review and publish community-contributed content, ensuring accuracy
- The phrasebook grows organically and stays relevant to real campus needs

1.4 Functional Requirements

- The system must support all eleven-official South African Languages
- The system must allow users to search phrases by keyword
- The system shall search across both English source text and all available translations simultaneously
- The system shall highlight the matching language in search results
- The system shall allow users to search or filter by category
- The system must allow users to search phrases by keyword
- The system must automatically log every time a phrase is viewed or searched
- The system must record which language was selected during each search/view
- The system must record which category the phrase belongs to

1.5 non-functional requirements

(Reliability & Availability)

- The system must be available to students and staff with minimal down-time
- The system must handle failed database connections by displaying a user-friendly error instead of crashing
- The system must operate 24/7 targeting an uptime of 99%
- The system should be resilient to concurrent usage during peak periods (e.g. registration week) without crashing

(Performance)

- Search results must return within 2 seconds
- The system must be able to handle at least 700 concurrent users without a noticeable slowdown

(Security)

- Passwords shall be securely hashed
- The system must enforce role-based access control
- The system shall protect against common web vulnerabilities

(Maintainability)

- The system shall follow a layered architecture (MVC pattern) to allow future developers to easily extend or modify features
- The codebase shall include clear naming conventions and documentation to support handover to future DUT IT staff or student teams

(Scalability)

- The system must be designed in a way that accommodates growth in phrases, languages, and user submissions over time,

User Requirements

3.1 User Authentication and Authorization

ID	Requirement	Priority	Actor
INNOFR-001	As a new user, I want to register with my email and password, so that I can create an account.	High	All
INNOFR-002	As a user, I want to log in and log out, so that I can access my account securely.	High	All
INNOFR-003	As a new user, I want to automatically be assigned the "Student" role, so that I have appropriate access from the start.	High	System
INNOFR-004	The system shall restrict access to administrative features to users with the "Administrator" role using [Authorize (Roles = "Administrator")] attributes at the controller level	High	System
INNOFR-005	The system shall allow unauthenticated users to search and view phrases (read-only access)	High	Guest

3.2 Phrase Repository Management

ID	Requirement	Priority	Actor
INNOFR-007	As a user, I want to access phrases in all 11 official South African languages, so that I can communicate in my preferred language.	High	All
ID	Requirement	Priority	Actor
INNOFR-008	As a user, I want phrases organized into categories, so that I can easily find what I need.	High	All
INNOFR-009	As an Administrator, I want to add new phrases with English text, a category, and translations, so that the phrasebook stays up to date.	High	Admin
INNOFR-010	As an Administrator, I want to edit existing approved phrases, so that I can fix errors or improve accuracy.	High	Admin
INNOFR-011	As an Administrator, I want to deactivate outdated phrases, so that users don't see irrelevant content.	Medium	Admin

INNOFR-012	As an Administrator, I want to add, edit, and deactivate categories, so that the phrasebook structure stays organized	Medium	Admin
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3.3 Multi-Language Search (Core Feature)

ID	Requirement	Priority	Actor
INNOFR-013	As a user, I want a prominent search bar on the home page, so that I can quickly start searching for phrases.	High	All
INNOFR-014	As a user, I want my search to check both the English text and all translations, so that I can find a phrase regardless of which language I search in.	High	All
INNOFR-015	As a user, I want autocomplete suggestions while I type, so that I can find phrases faster.	High	All
INNOFR-016	As a user, I want the matching text highlighted in the results, so that I can quickly see why a result matched my search.	High	All
INNOFR-017	As a user, I want to filter search results by category, so that I can narrow down results to what's relevant to me.	High	All
INNOFR-018	As a user, I want each result to show the English phrase, translation, language, and category, so that I have full context immediately.	High	All
INNOFR-019	As a user, I want a helpful message when no results are found, so that I know what to try next.	Medium	All
ID	Requirement	Priority	Actor
INNOFR-020	As a user, I want search results to load without a full page refresh, so that my experience feels fast and seamless.	Medium	All

3.4 Translation Submission Workflow (Community Contribution)

ID	Requirement	Priority	Actor
INNOFR-021	As a logged-in Student, I want to submit a translation suggestion for an existing phrase, so that I can contribute to the phrasebook's content.	High	Student
INNOFR-022	As a Student, I want to select the target language for my translation from a dropdown of all 11 languages, so that I can specify which language my suggestion is for.	High	Student
INNOFR-023	As a Student, I want my submitted translation to be stored with a "Pending" status, so that it can be reviewed before being made public.	High	System
INNOFR-024	As a user, I want "Pending" translations to be excluded from search results, so that I only see verified, approved content.	High	System
INNOFR-025	As an Administrator, I want to view all pending translation submissions in a dashboard with Approve/Reject buttons, so that I can efficiently review contributions.	High	Admin
INNOFR-026	As an Administrator, I want to approve a pending translation, so that it becomes visible to users in search results.	High	Admin
INNOFR-027	As an Administrator, I want to reject a pending translation with a reason, so that the student understands why it wasn't accepted.	High	Admin
INNOFR-028	As an Administrator, I want to add audio pronunciation to a phrase, so that users can learn how to correctly pronounce it.	High	Admin
ID	Requirement	Priority	Actor
INNOFR-029	As a Student, I want to request a correction from an Administrator on a phrase I submitted, so that inaccurate translations can be reviewed and fixed.	Medium	Student
INNOFR-030	As a Student, I want to see the approval status of my submitted translations on my profile page, so that I can track the progress of my contributions.	Medium	Student

INNOFR-031	As a user, I want to see a randomly generated phrase of the day on the home page, so that I can discover new phrases and stay engaged with the app.	Medium	System
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3.5 Favourites System

ID	Requirement	Priority	Actor
INNOFR-032	As a logged-in Student, I want to save a phrase to my favourites list, so that I can quickly access phrases I use often.	High	Student
INNOFR-033	As a Student, I want to remove a phrase from my favourites list, so that I can keep my list relevant to my current needs.	High	Student
INNOFR-034	As a Student, I want a "My Favourites" page showing all my saved phrases, so that I can easily review and revisit them.	High	Student
INNOFR-035	As a Student, I want my favourite phrases to be reliably saved and linked to my account, so that my list persists across sessions.	High	System
INNOFR-036	As a Student, I want to be prevented from adding the same phrase to my favourites twice, so that my list stays clean and accurate.	Medium	System

3.6 Language Usage Statistics (Analytics)

ID	Requirement	Priority	Actor
INNOFR-037	As an Administrator, I want the system to automatically log every phrase view, so that I can understand which phrases are most used.	Medium	System
INNOFR-038	As an Administrator, I want the system to automatically log every search query, so that I can analyze user search behavior.	Medium	System
ID	Requirement	Priority	Actor
INNOFR-039	As an Administrator, I want the system to log which language was used in each search or view, so that I can identify language usage trends.	Medium	System
INNOFR-040	As an Administrator, I want the system to log which category each viewed phrase belongs to, so that I can understand which categories are most accessed.	Medium	System
INNOFR-041	As an Administrator, I want to view a report showing which languages are searched most frequently, so that I can prioritize content development for high-demand languages.	Medium	Admin

INNOFR-042	As an Administrator, I want to view a report showing which categories are accessed most frequently, so that I can focus improvement efforts on the most-used categories.	Medium	Admin
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3.7 Student Profile

ID	Requirement	Priority	Actor
INNOFR-043	As a Student, I want a profile page, so that I can view my personal activity within the system.	Medium	Student
INNOFR-044	The profile page shall display all phrases the Student has submitted, with their current approval status (Pending, Approved, Rejected) As a Student, I want to see all phrases I've submitted along with their approval status (Pending, Approved, Rejected), so that I can track the outcome of my contributions.	Medium	Student
INNOFR-045	The profile page shall display the Student's saved favourites count As a Student, I want to see the number of phrases I've saved as favourites on my profile, so that I can quickly gauge my engagement with the app.	Low	Student
INNOFR-046	The profile page shall display the Student's total number of approved contributions As a Student, I want to see my total number of approved contributions, so that I can track my impact on the phrasebook.	Low	Student

3.8 Administrator Dashboard

ID	Requirement	Priority	Actor
INNOFR-047	As an Administrator, I want a dashboard with links to all administrative functions, so that I can efficiently manage the system from one place.	High	Admin
INNOFR-048	As an Administrator, I want a dashboard with links to all administrative functions, so that I can efficiently manage the system from one place.	High	Admin
INNOFR-049	As an Administrator, I want to see recent activity such as new submissions and approvals on the dashboard, so that I can stay informed about ongoing system usage.	Low	Admin

Use Case Descriptions

Use case number	01
Use case name	<i>Register account</i>
Brief description	<i>A new user creates an account to access student specific features</i>
Triggering event	User wants to have access to the system features
Actor(s)	Student
Preconditions	<i>User is not logged in</i>
Flow of activities	1. <i>User navigates to registration page</i> 2. <i>User enters email, password, and confirm password.</i> 3. <i>System validates input (email format, password strength)</i> 4. <i>System creates new user account with role "Student"</i> 5. <i>System redirects user to Home page</i>

Use case number	02
Use case name	<i>Search phrase</i>
Brief description	<i>A student search for a phrase by entering the keyword or phrase and selecting their preferred language and category</i>
Triggering event	User wants to translate a phrase
Actor(s)	Student
Preconditions	<i>User must be login in</i>
Flow of activities	6. <i>Student login to the system</i> 7. <i>System validates login credentials</i> 8. <i>Student enters keyword(s) or selects a category</i> 9. <i>Student selects preferred language</i> 10. <i>Student searches for the phrase</i> 11. <i>System searches across English text and all translations</i> 12. <i>System displays results with matching highlighted text</i>

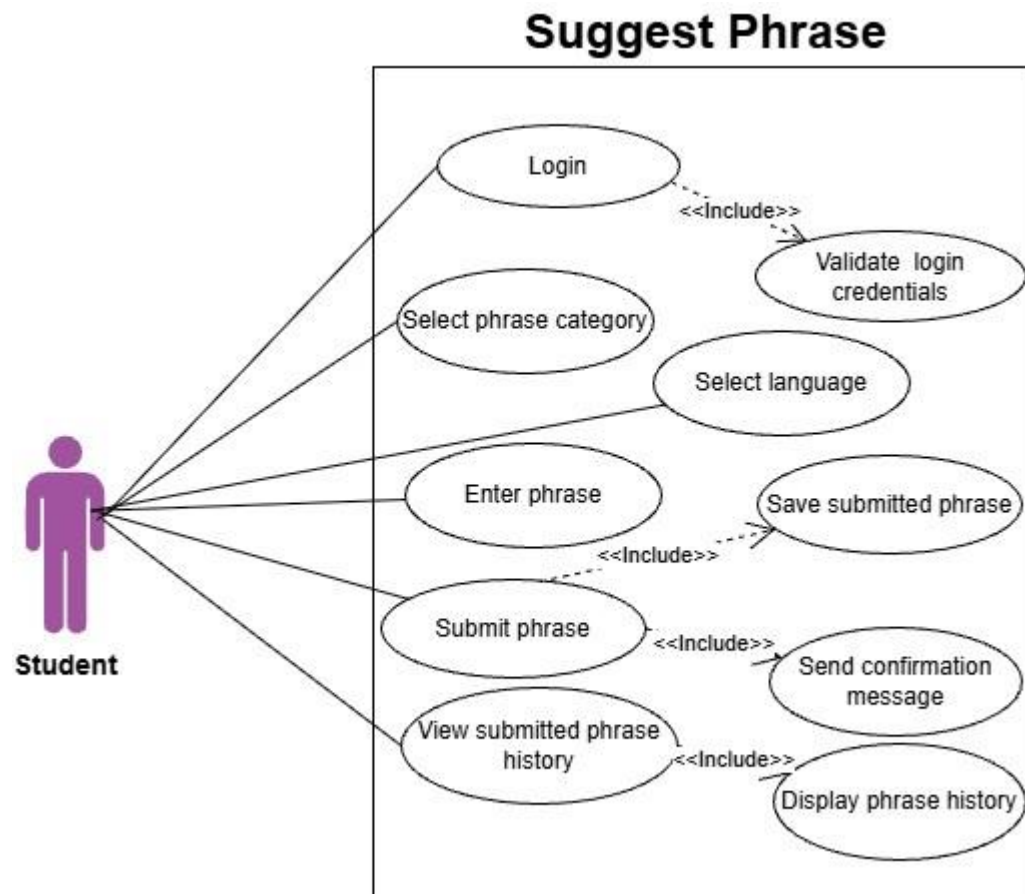
Use case number	03
Use case name	<i>Submit phrase translation</i>
Brief description	<i>A student requests to make a phrase translation, by choosing the language, the category the phrase belongs, and entering and submitting the phrase. Which the status goes to pending until reviewed by the admin.</i>
Triggering event	Student desires to submit new phrase translation
Actor(s)	Student
Preconditions	<i>User must be login in</i>

Flow of activities	13. Student login to the system 14. Student clicks "Suggest translation" button 15. Student selects phrase category 16. Student selects preferred language and enters translation text 17. Student submits the request 18. System validates translation (Language not already translated, text empty) 19. System saves translation status with "Pending" 20. System displays confirmation message
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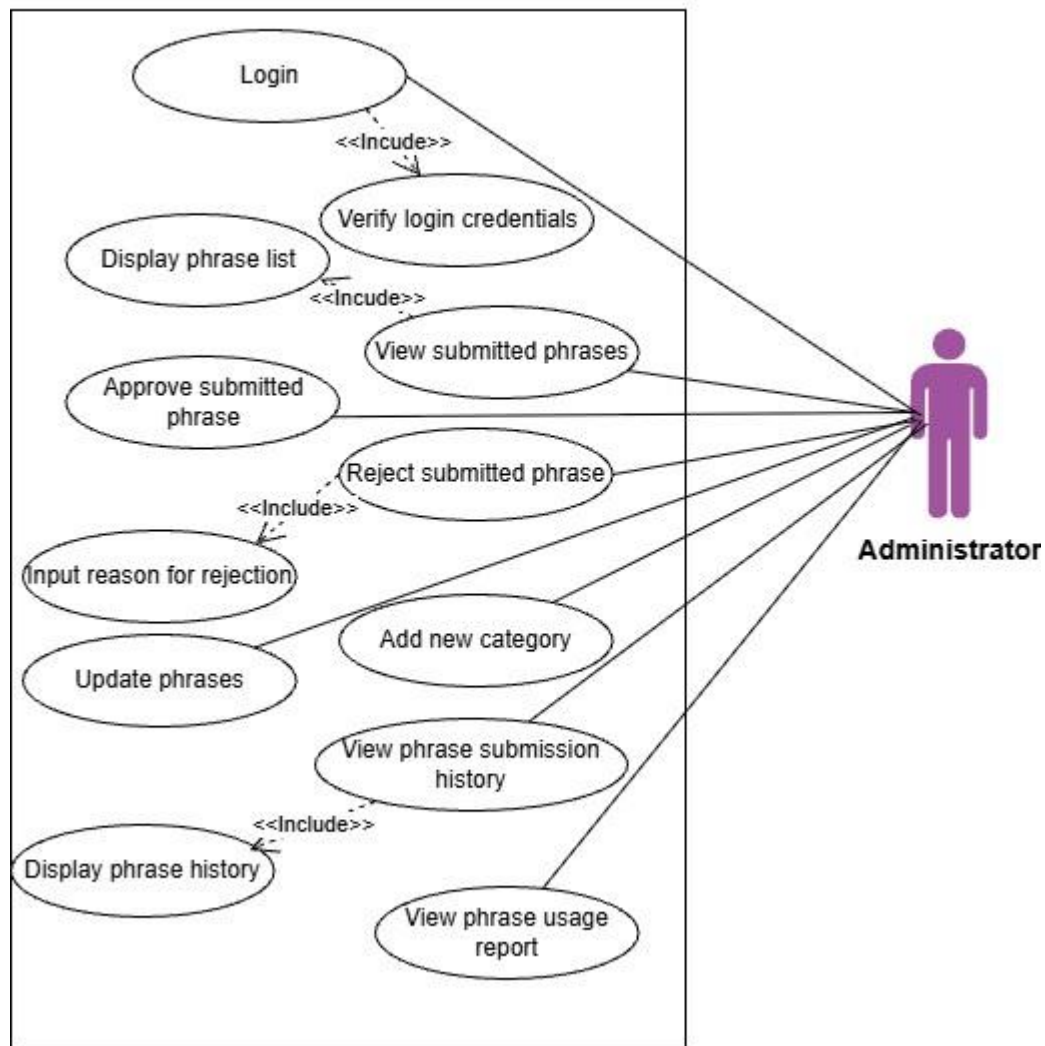
Use case number	04
Use case name	Manage submit phrase
Brief description	Administrator manages submitted phrases, approves or reject with feedback, and adds new categories and language,
Triggering event	Students submitted phrase translation suggestions
Actor(s)	Admin
Preconditions	User must be login in
Flow of activities	21. Admin logs in to the system 22. Student clicks "View submitted phrases" 23. Admin review submitted phrases 24. System displays all submitted phrases list 25. Admin Rejects phrases 26. System prompts admin for rejection reason 27. Admin Enters rejection reason 28. System sends message to student 29. System changes students' status from "Pending" to "Rejected"

Use case number	05
Use case name	Update Phrase
Brief description	Administrator adds a new phrase (English text) to the phrasebook.
Triggering event	3a. Phrase already exists → System displays warning.
Actor(s)	Admiin
Preconditions	Administrator is logged in. Categories exist.
Flow of activities	1. Administrator navigates to Admin → Manage Phrases. 2. Administrator clicks "Add Phrase". 3. Administrator enters English text and selects category. 4. Administrator submits the form. 5. System validates input (text not empty, category exists). 6. System saves the new phrase.

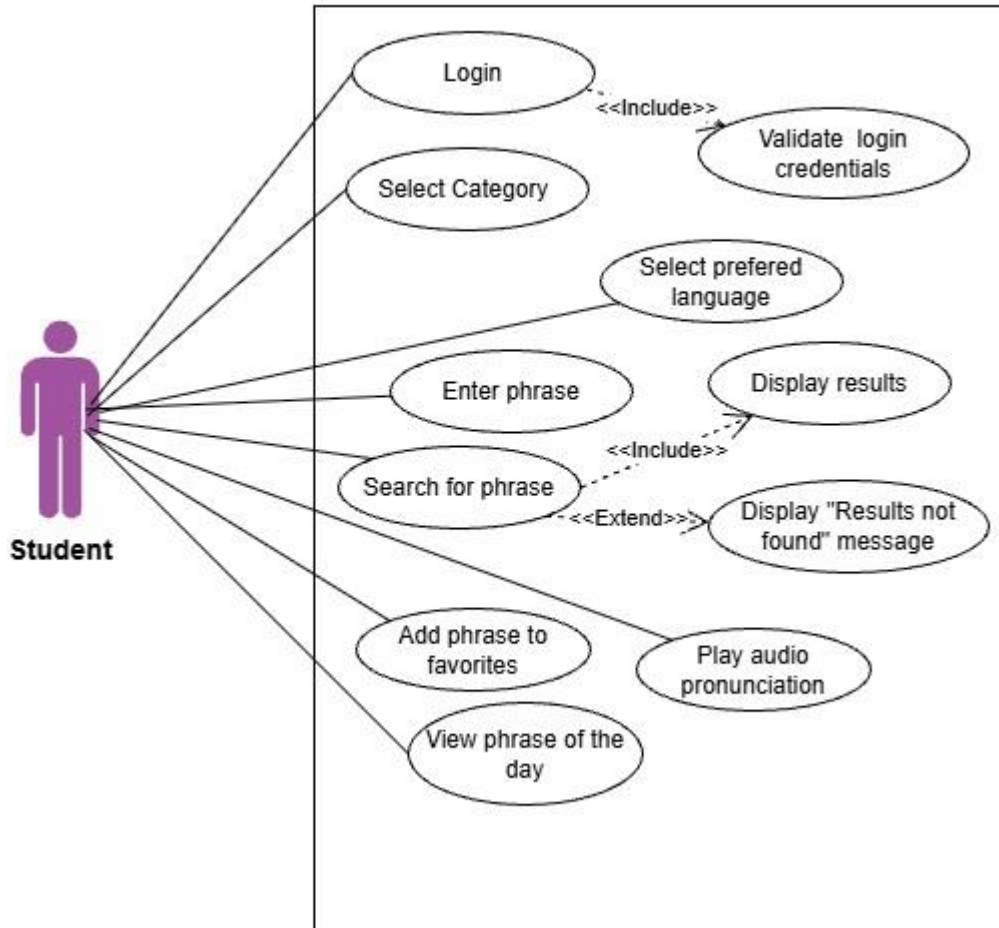
Use Case Diagrams



Manage Submitted Phrases



Search Phrase



Wireframes

Home page

 **Explore phrases**

Try Student or Admin dashboards

★ Phrase of the day

"Umntu ngumuntu ngabantu"

 Listen  Philosophy

A person is a person through other people

 Trending: Sawubona

 Categories: 12

 Languages: 8

 Favorites: 3.2k

Search for a phrase... (try 'Sawubo

All categories ▼

All languages ▼

Search

★ **Sawubona**

isiZulu → English: "Hello"

👁 1.2k searches  Listen

Save

Suggest correction

Student Dashboard

15

Saved phrases

8

Pending suggestions

3

Rejected

100+

Total searches

Search for a phrase...

Category 

Target language 

 Search

 Suggest phrase

 Favorites

 History

 Recent activity

 **Sawubona → isiZulu**

Approved • 2 days ago • 50 searches

Approved

 **Ngiyabonga → isiZulu**

Pending review • 5 days ago • 30 searches

Pending

Admin Dashboard

2,456

Total phrases

156

Pending suggestions

23

Rejected today

1,234

Approved today

 Today's searches: **8,947**  Most searched: **Sawubona** (342)  Active: **1,234**

 New suggestions: **23**

 Pending reviews (12)

Unjani → isiZulu

User42 • 2 hrs ago

Pending

Ngiyabonga → Sesotho

User15 • 3 hrs ago

Pending

 Quick manage

 Manage categories & languages

 View analytics

 Add audio pronunciation